Velocity gradient prediction using parameterized Lagrangian deformation models

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Abstract

We seek to efficiently predict the statistical evolution of the velocity gradient tensor (VGT) by creating local models for the pressure Hessian. Previous work has identified physics-informed machine learning (PIML) to be adept in this prediction; of note in this class of models is the Tensor Basis Neural Network (TBNN) for its embedded physical constraints and demonstrated performance. Simultaneously, phenomenological models were advanced by approximating the local closure to the pressure Hessian via deformation models using the history of the VGT. The latest in this series of models is the Recent Deformation of Gaussian Fields (RDGF) model. In this work, we combine the (local in time) PIML approach with the phenomenological idea of Lagrangian deformation to create a data-driven Lagrangian deformation model. We compare the model performance to both the TBNN and the RDGF models, and provide data-driven hypotheses regarding the Lagrangian upstream assumptions made in the RDGF model.

1 Introduction

The Lagrangian velocity gradient tensor (VGT) captures many fundamental aspects of turbulence; it displays non-Gaussian statistics (e.g., intermittency) including those of energy dissipation, describes the deformation rate of a fluid volume, and encapsulates alignment between strain and vorticity. Because of its fundamental nature, understanding the evolution of the VGT is central to modern turbulent modeling challenges. E.g., creating better sub-grid models in LES[23]. However, despite ever-growing computation capability, resolving the VGT using direct numerical simulation (DNS) of Navier-Stokes is still prohibitively expensive for realistic Reynolds numbers.

Initiated by the work of Viellefosse, who examined keeping only the purely local contribution to the VGT evolution, "local closures" for the evolution of the VGT have recieved considerable attention. Viellefosse and subsequent work from Cantwell showed a finite time singularity originating from this so-called Restricted Euler dynamics. Further development in these local closures worked to resolve this finite time singularity; e.g., stochastic diffusion model[9], linear damping model [16], tetrad model [4], linear diffusion model[11], recent fluid deformation approximation [6], among others[17].

Throughout this development, analysis of DNS datasets have motivated and tested model hypotheses. This hisory of data-driven synthesis dovetails naturally with advances in machine learning (ML). While ML has been applied liberally to turbulence in the Eularian frame, see e.g., [8] [19], only recently has it been applied to Lagrangian turbulence, [1][15].

Building on the phenomenological ideas of Lagrangian deformation, and parameterizing these contributions using PIML, we create a parameterized Lagrangian deformation model (PLD). The rest of the manuscript is organized as follows: in section 2 we present the two driving approaches and establish recent state-of-the-art, in section 3 we present the PLD model, its connection to previous Lagrangian deformation approaches, and finally present results in 4.

2 Background

Throughout, we denote scalar fields with lower-case Latin letters; vector fields with subscripted, lower-case Latin letters; 2nd-rank tensors with upper-case Latin letters; and constants using lower-case Greek alphabet. The two exceptions to this rule are the second and third scalar invariants of the velocity gradient tensor, denoted by Q, R respectively, which we maintain in congruence with the literature.

For all tensors in the paper, indices will be denoted by one or more of $i, j, k, l, m, n \in [1, 2, 3]$. We follow the Einstein summation notation, that repeated indices imply summation, e.g. $A_{ii} = \mathrm{tr}(A)$, $u_k \frac{\partial u_i}{\partial x_k} = u \cdot \nabla u$, $\frac{\partial^2 u_i}{\partial x_k \partial x_k} = \nabla^2 u$, etc.

2.1 Governing Equations for VGT

Incompressible Navier-Stokes defines the evolution of a velocity field $u(x, t)$, given a pressure field $p(x, t)$

$$\frac{\partial u_i}{\partial t} + u_k \frac{\partial u_i}{\partial x_k} = -\frac{\partial p}{\partial x_i} + \nu \frac{\partial^2 u_i}{\partial x_k \partial x_k} \tag{1}$$

$$\frac{\partial u_i}{\partial x_i} = 0 \tag{2}$$

Appling spatial derivatives to eq(1), and using the definition of material derivative, we obtain an ODE for the Lagrangian VGT:

$$\frac{dA_{ij}}{dt} = \frac{\partial A_{ij}}{\partial t} + u_k \frac{\partial A_{ij}}{\partial x_k} = -A_{ik}A_{kj} - \frac{\partial^2 p}{\partial x_i \partial x_j} + \nu \frac{\partial^2 A_{ij}}{\partial x_k \partial x_k} \tag{3}$$

Taking the trace of eq(3) and using the incompressibility condition eq(2), we determine the trace of the pressure Hessian

$$\frac{\partial^2 p}{\partial x_k \partial x_k} = -A_{ij}A_{ji} \tag{4}$$

Thus, we write the evolution equation for the incompressible, Lagrangian VGT as

$$\frac{dA_{ij}}{dt} = -\left(A_{ik}A_{kj} - \frac{1}{3}A_{mn}A_{nm}\delta_{ij}\right) - \left(\frac{\partial^2 p}{\partial x_i \partial x_j} - \frac{1}{3}\frac{\partial p}{\partial x_k \partial x_k}\delta_{ij}\right) + \nu \frac{\partial^2 A_{ij}}{\partial x_k \partial x_k} \tag{5}$$

This has three clear contributions: the purely local, "Restricted Euler" (RE) term

$$E_{ij} := -\left(A_{ik}A_{kj} - \frac{1}{3}A_{mn}A_{nm}\delta_{ij}\right) \tag{6}$$

the deviatoric (or nonlocal) pressure Hessian

$$H_{ij} := -\left(\frac{\partial^2 p}{\partial x_i \partial x_j} - \frac{1}{3}\frac{\partial p}{\partial x_k \partial x_k}\delta_{ij}\right) \tag{7}$$

and the viscous contributions

$$T_{ij} := \nu \frac{\partial^2 A_{ij}}{\partial x_k \partial x_k} \tag{8}$$

Using this simplified notation, the ODE for the Lagrangian VGT takes the form

$$\frac{dA_{ij}}{dt} = E_{ij} + H_{ij} + T_{ij} \tag{9}$$

2.2 Local Closures for the Evolution Equations

While the viscous term is dynamically important to damp turbulent fluctuations, numerical studies confirm that the leading order (nonlocal) contribution to the VGT evolution is the deviatoric pressure Hessian[16][25]. Therefore, in this section, we review the literature from the perspective of closing this term; we should note however, that historically the deviatoric pressure Hessian and viscous terms were treated together.

2.2.1 Lagrangian Deformation Models

In the first attempt to locally close the VGT equations, Viellefosse [26] studied the RE dynamics:

$$\frac{dA_{ij}}{dt} \approx E_{ij} \tag{10}$$

His asymptotic analysis found this reduction induced a finite-time singularity. This was confirmed by subsequent investigation by Cantwell et.al[2][3], who specified the exact solution and studied invariant dynamics and subsequently topological evolution in RE dynamics. Later, [14] directly studied material deformation under RE dynamics and confirmed the asymptotic flattening of fluid elements, corresponding to an acceleration down the "Viellefosse tail" in the Q-R phase plane.

The universal behavior of fluid elements was one motivation for Chertkov et.al to introduce the Tetrad Model[4] that used 4 points to track and accumulate the history of the strain on a fluid element. This history was then utilized to directly oppose the extreme flattening of the element and the acceleration of the phase plane dynamics along the Viellefosse tail. The tetrad model was interpreted and - using a short-history approximation - adapted into the recent fluid deformation (RFD) approximation [6]. Further development of this idea enriched the upstream initial condition to a Gaussian field approximation.

Johnson and Meneveau [12] introduced the recent deformation of Gaussian fields (RDGF) model following a long line of postulating that the deviatoric part of the pressure Hessian could be locally modeled using information of the time history of the deformation tensor [4],[6],[5].

Johnson and Meneveau [12] introduced the recent deformation of Gaussian fields (RDGF) model, following this line of postulation, and enriching the upstream condition via fitting coefficients of a tensor expansion introduced originally by Pope[22]. The tensor expansion is treated in detail in section 2.2.2.

RDGF models the conditional average of the (full) pressure Hessian tensor as:

$$\hat{P}_{ij} = \frac{\partial y_k}{\partial x_i} \frac{\partial^2 p}{\partial y_k \partial y_l} \frac{\partial y_l}{\partial x_j} = D^{-1}_{ki} \tilde{P}_{kl} D^{-1}_{lj}, \tag{11}$$

where $\tilde{P}$ is the pressure Hessian tensor/matrix evaluated along the Lagrangian trajectory upstream in time. The deformation tensor, defined as a sensitivity of a particle observed at the position x at the moment of time t to its initial position y at the previous moment of time,

$$D_{ij} = \frac{\partial x_i}{\partial y_j}, \tag{12}$$

evolves according to

$$\frac{dD_{ij}}{dt} = A_{ik} D_{kj}, \quad \text{with } D_{ij}(0) = \delta_{ij}, \tag{13}$$

assuming isotropic initial condition (without loss of generality for sufficiently far removed upstream times). Solution of Eq. (13) is defined via the time-ordered exponential

$$D(t) = \text{Texp}\left(\int_0^t dt' A(t') \right) = \lim_{N\to\infty} \prod_{i=0}^{N} \left(e^{A(t_i)\Delta t} \right), \tag{14}$$

where $t_i = i\Delta t$ and $\Delta t = t/N$. If we take $N = 1$ and set Δt to be the smallest temporal scale of turbulence (also called the Kolmogorov, or viscous scale) we arrive at the RDGF approximation for the deformation tensor

$$\hat{D}(x,t) = \exp\left(A(x,t)\Delta t\right) \approx D(\Delta t). \tag{15}$$

All that is left is to determine the upstream pressure Hessian $\tilde{P}$. The RDGF model uses a Gaussian field closure for $\tilde{P}$,

$$\hat{\tilde{P}}_{ij} = \frac{1}{3}\tilde{P}_{kk}\delta_{ij} + \langle \tilde{P}^{(d)}_{ij} | A \rangle_{\text{Gaussian}} \tag{16}$$

where $\langle \tilde{P}^{(d)} | A \rangle_{Gaussian}$ is the mean deviatoric pressure Hessian in a Gaussian field, found by Wilczek & Meneveau[27]. Using the notation of the tensor basis [Eqs. (27-31)], they found

$$\langle \tilde{P}^{(d)}_{ij} | A \rangle_{\text{Gaussian}} = \gamma T^{(2)} + \alpha T^{(3)} + \beta T^{(4)} \tag{17}$$

with the value of the coefficients found from a mix of theoretical and numerical experiments

$$\alpha = -\frac{2}{7}, \quad \beta = -\frac{2}{5}, \quad \gamma \approx 0.08. \tag{18}$$

Introducing the deformed Gaussian field approximation

$$\hat{G}_{ij} := \hat{D}^{-1}_{mi} \langle \tilde{P}^{(d)}_{mn} | A \rangle_{\text{Gaussian}} \hat{D}^{-1}_{nj}, \tag{19}$$

and a short-time approximation to the inverse of the left Cauchy-Green tensor

$$\hat{C}^{-1}_{ij} = \hat{D}^{-1}_{ki}\hat{D}^{-1}_{kj} \tag{20}$$

We enforce the prediction in Eq. (16) to have the correct trace, and arrive at the following RDGF prediction for the component of the pressure Hessian conditioned to the VGT

$$\hat{P}_{ij} = 2Q\frac{\hat{C}^{-1}_{ij}}{\hat{C}^{-1}_{kk}} + \hat{G}_{ij} - \frac{\hat{C}^{-1}_{ij}}{\hat{C}^{-1}_{kk}}\hat{G}_{ll}. \tag{21}$$

2.2.2 Tensor Basis Neural Network

Contemporaneous to the development of the RDGF model, work was being done to use tensor expansion approximations to the pressure Hessian. Leveraging previous work from Pope[22], Lawson and Dawson fit coefficients to a truncated tensor expansion using DNS data[13]. Tian et.al enriched this approach by parameterizing the coefficient functions, resulting in a so-called tensor basis neural network[25].

In the case of incompressible turbulence we can write the formal (nonlocal) solution for the deviatoric pressure Hessian as an integral over the spatial domain[18]

$$H_{ij}(\boldsymbol{x}) = \iiint \frac{\delta_{ij} - \hat{r_i}\hat{r_j}}{2\pi r^3} Q(\boldsymbol{x}+\boldsymbol{r})d\boldsymbol{r} \tag{22}$$

where $Q = \frac{1}{2}\mathrm{tr}(A^2)$, and $\boldsymbol{x} \in \mathbb{R}^3$.

Lawson & Dawson [13], proposed a local closure based an expansion of the integral in eq(22); first via a Taylor series expansion in words of the strain and rotation rate tensors,

$$\hat{H} = \sum_{m,n=0}^{\infty} \alpha_{mn} S^m W^n \tag{23}$$

$$S = \frac{1}{2}(A + A^T) \qquad W = \frac{1}{2}(A - A^T) \tag{24}$$

Then reduced the infinite sum using Cayley-Hamilton theorem on A, and expanded via the tensor basis for traceless, symmetric, rotationally invariant function[28],[21]

$$\hat{H} = \sum_{n=1}^{10} g^{(n)}(\lambda_1, \ldots, \lambda_5) T^{(n)} \tag{25}$$

with $g^{(n)}$ scalar functions of the invariants

$$\lambda_1 = \mathrm{tr}(S^2) \quad \lambda_2 = \mathrm{tr}(W^2) \quad \lambda_3 = \mathrm{tr}(S^3) \quad \lambda_4 = \mathrm{tr}(W^2 S) \quad \lambda_5 = \mathrm{tr}(W^2 S^2) \tag{26}$$

and the tensor basis given by:

$$T^{(1)} = S \qquad T^{(2)} = SW - WS \tag{27}$$

$$T^{(3)} = S^2 - \frac{1}{3}I \cdot \mathrm{tr}(S^2) \qquad T^{(4)} = W^2 - \frac{1}{3}I \cdot \mathrm{tr}(W^2) \tag{28}$$

$$T^{(5)} = WS^2 - S^2W \qquad T^{(6)} = W^2S + SW^2 - \frac{2}{3}I \cdot \mathrm{tr}(SW^2) \tag{29}$$

$$T^{(7)} = WSW^2 - W^2SW \qquad T^{(8)} = SWS^2 - S^2WS \tag{30}$$

$$T^{(9)} = W^2S^2 + S^2W^2 - \frac{2}{3}I \cdot \mathrm{tr}(S^2W^2) \quad T^{(10)} = WS^2W^2 - W^2S^2W \tag{31}$$

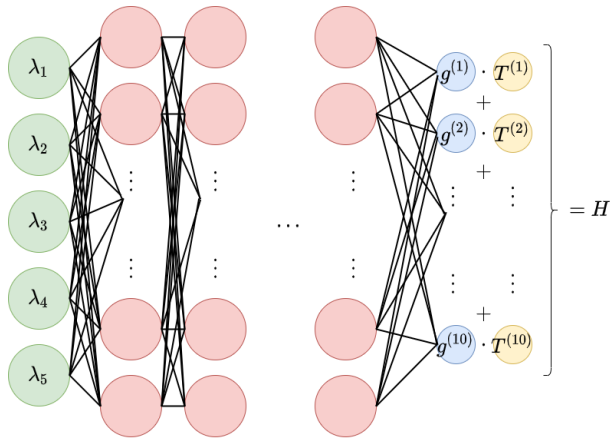

Figure 1: The architecture of the TBNN. The invariants and tensor basis elements are calculated from each sample of the VGT, the invariants are then used as input to a fully connected network, and the resulting $g^{(i)}$ multiply the corresponding tensor basis elements $T^{(i)}$, before summing into the prediction $\hat{H}$

This formulation reduces the challenge to finding the functions of known scalars. Lawson and Dawson fit constants to a truncated basis expansion (as did the RDGF model for their upstream condition). Tian et.al enriched this approach by learning the $g^{(n)}$'s as shown in fig(1) and using the entire tensor basis. In particular, this allowed the scalar functions to be nontrivial functions of the invariants.

The TBNN architecture ensures the output tensor $\hat{H}$ is symmetric, traceless, and the network itself is rotationally and Galilean invariant. A technical note, since we are using the L_2 loss, $\hat{H}$ is only an approximation to the mean deviatoric pressure Hessian if the statistics of the pressure Hessian are normal. As we expect turbulent statistics to be non-Gaussian, the choice of loss is driven more by mathematical convenience.

Tian et.al [25] showed ability to train the network, and demonstrated state-of-the-art performance on a variety of relevant physical metrics: eigenvector alignments, Q-R conditional mean tangents (CMTs), and *a posteriori* tests evolving an initially Gaussian field to fully developed turbulence as evidenced by characteristic teardrop-shaped Q-R probability distributions (PDFs).

3 Methodology

Our approach biases the statistics of the conditional deviatoric pressure Hessian using the history of the VGT (as in the RDGF, RFD, and Tetrad models), while avoiding postulating the exact functional form; instead allowing it to be data-driven using PIML (as in TBNN model).

We do this by augmenting the inputs to the feed-forward portion of the TBNN, the idea being to bias the statistics of the invariants, while believing that the tensors derived from the local-in-time VGT are sufficient to span the function space.

3.1 Temporal Convolution

Motivated by the deformation tensor in eq(14)

$$D(t) = \text{Texp}\left(\int_0^t dt' A(t')\right) = \lim_{N\to\infty} \prod_{i=0}^{N} \left(e^{A(t_i)\Delta t}\right), \tag{32}$$

Each term on the right-hand side of eq(14) can be expanded to

$$e^{A_m \Delta t} = I + \sum_{i=0}^{N} \frac{1}{i!} \left(\Delta t A_m\right)^i = I + \Delta t A_m + \mathcal{O}(\Delta t^2) \tag{33}$$

So that, taking the leading order in Δt, we obtain the first-order approximation of the deformation tensor

$$D = I + \Delta t \sum_{n=N}^{0} A_n + \mathcal{O}(\Delta t^2) \tag{34}$$

Therefore, the simplest of our proposed models augments the set of invariants using a learned temporal convolution

$$\hat{H} = \sum_{n=1}^{10} g_\theta^{(n)}(\lambda_1, \ldots, \lambda_5, C) T^{(n)} \tag{35}$$

with

$$C(t) = \sum_{t_f}^{t_f - \Delta t \cdot n} \zeta_n A_n \tag{36}$$

where the scalar ζ_n are learned simultaneously with the parameterization of the scalar g functions.

This model avoids prejudicing the prediction with a specific phenomenological theory of history importance, at the cost of increased computational overhead (namely storing history for each VGT). However, because of the simple structure, the learned ζ's may inform a temporal correlation in the VGT that is useful in the prediction of the deviatoric pressure Hessian. An extreme case of very short time correlations would reinforce the hypothesis presented in RDGF, while longer temporal correlations would support longer memory approaches e.g., [24].

3.2 Time-Ordered Exponential

Further, we may hypothesize the exact form of history, replacing C in Eq. (35) by the true time-ordered exponential

$$W(t) = \text{Texp}\left(\int_0^t dt' A(t')\right) \tag{37}$$

3.3 Ground Truth Data & Training

Our data is DNS of forced isotropic turbulence with $Re \approx 240$. Eularian data is generated, random points are sampled, VGT and pressure Hessian are constructed at these points using interpolation, and Lagrangian trajectories of these quantities are advanced using the Eularian velocity fields. We have 122k samples, each advanced for 1000 timesteps with $\Delta t = 3e\text{-}4$, representing approximately an inertial eddy turnover time. For each sample and each timestep, we have measured VGT, PH, and viscous terms. We use single timesteps, aggregated over Lagrangian paths separated in time by half an eddy turnover time, as labeled data. As observed by Parashar et.al [20], the behavior and training of the TBNN is sensitive to choice of normalization. We follow Tian et.al choosing the physically inspired normalization by Kolmogorov timescale to normalize our input data (and thus the tensor basis itself)

$$\tau = \frac{1}{\langle \|S_{ij}\|_2^2 \rangle} \tag{38}$$

However, recent work suggests this may be an additional hyperparameter to consider[10].

We use a network architecture following Tian et.al's suggestion in their TBNN work[25], refined using a hyperparameter search[7] across network sizes and shapes, learning rates, and minibatch sizes for each augmentation to the TBNN. The network is trained to minimize the L_2 error between the ground truth and predicted pressure Hessian. That is, the optimal parameters are the solution to the optimization

$$\min_{\theta} \left\| H - \hat{H} \right\|_2^2 \tag{39}$$

4 Results

To evaluate performance of the models, we rely initially on validation loss values as a coarse measure of out-of-testset performance; historically consulted Q-R phase plane conditional mean tangents (CMTs) as a sanity check; then pressure Hessian eigenvector alignment PDFs as a richer measure of meaningful performance; and finally *a posteriori* tests of Q-R phase plane PDFs and VGT component PDFs. Comparison of minimum validation loss values tend to indicate general performance across this set of measures, while the remaining measures are in order of increasing physical significance.

5 Conclusion

In this work (in progress), we reproduced the current state-of-the-art Lagrangian model to predict the PH, contextualized it with recent phenomenological advancements, and advanced the methodology by introducing and evaluating a new predictive model.

The TBNN is a powerful tool to capture symmetries, while allowing for the flexibility of a NN. We applied this model to homogeneous, isotropic turbulence to close the equations of the non-local pressure Hessian using only the local velocity gradient tensor. We contrasted this approach with a mathematically similar, but philosophically different approach using the RDGF model, motivated by the idea that the recent history of the VGT could better inform the prediction of the current PH.

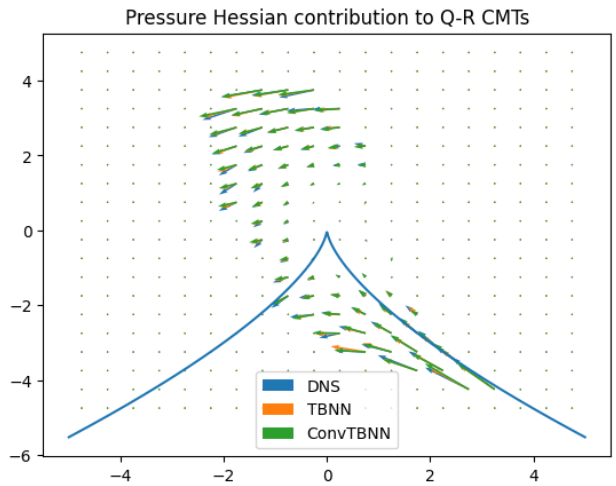

Figure 2: Qualitatively, the Q-R conditional mean tangents match the performance of the TBNN

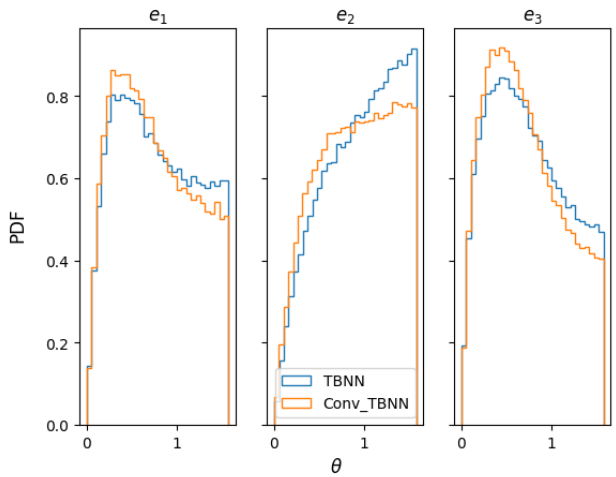

Figure 3: The ConvTBNN outperforms the TBNN in pressure Hessian eigenvector alignment PDF

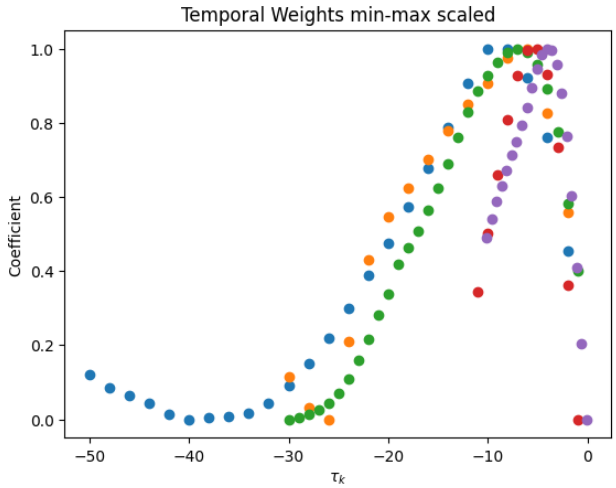

Figure 4: Note 3 items: 1) The convolution weights are small in far history, and for short kernels no flat tails are observed; 2) Weights are small for very recent history; 3) Weights in some importance region increase monotonically.

By combining the two, we created the convTBNN, a first data-driven step towards biasing the statistics of the current PH prediction using the history of the VGT. We showed that we could outperform the local in time TBNN in two metrics, loss value and eigenvector alignment. We began to interpret the weights of the temporal convolution kernel, and related them to the hypotheses made in the phenomenological models (RFD, RDGF).

Further work needs to be done to ensure these results hold under evaluation of richer metrics such as the Q-R conditional mean tangents, and evaluate the accuracy and stability of the resulting differential equation for the VGT.

References

[1] Dhawal Buaria and Katepalli R. Sreenivasan. Forecasting small-scale dynamics of fluid turbulence using deep neural networks. *Proceedings of the National Academy of Sciences*, 120(30), July 2023.

[2] Brian J. Cantwell. Exact solution of a restricted euler equation for the velocity gradient tensor. *Physics of Fluids A: Fluid Dynamics*, 4(4):782–793, 1992.

[3] Brian J Cantwell. On the behavior of velocity gradient tensor invariants in direct numerical simulations of turbulence. *Physics of Fluids A: Fluid Dynamics*, 5(8):2008–2013, 1993.

[4] Michael Chertkov, Alain Pumir, and Boris I. Shraiman. Lagrangian tetrad dynamics and the phenomenology of turbulence. *Physics of Fluids*, 11(8):2394–2410, 1999.

[5] L Chevillard, C Meneveau, L Biferale, and F Toschi. Modeling the pressure hessian and viscous laplacian in turbulence: comparisons with direct numerical simulation and implications on velocity gradient dynamics. *Physics of Fluids*, 20(10), 2008.

[6] Laurent Chevillard and Charles Meneveau. Lagrangian dynamics and statistical geometric structure of turbulence. *Physical review letters*, 97(17):174501, 2006.

[7] François Chollet et al. Keras. `https://keras.io`, 2015.

[8] Karthik Duraisamy, Gianluca Iaccarino, and Heng Xiao. Turbulence modeling in the age of data. *Annual review of fluid mechanics*, 51:357–377, 2019.

[9] SS Girimaji and SB Pope. A diffusion model for velocity gradients in turbulence. *Physics of Fluids A: Fluid Dynamics*, 2(2):242–256, 1990.

[10] Criston Hyett, Yifeng Tian, Michael Woodward, Michael Chertkov, Daniel Livescu, and Mikhail Stepanov. Applicability of machine learning methodologies to model the statistical evolution of the coarse-grained velocity gradient tensor. *Bulletin of the American Physical Society*, 2022.

[11] Eunhwan Jeong and Sharath S Girimaji. Velocity-gradient dynamics in turbulence: effect of viscosity and forcing. *Theoretical and computational fluid dynamics*, 16(6):421–432, 2003.

[12] Perry L. Johnson and Charles Meneveau. A closure for lagrangian velocity gradient evolution in turbulence using recent-deformation mapping of initially gaussian fields. *Journal of Fluid Mechanics*, 804:387–419, 2016.

[13] J. M. Lawson and J. R. Dawson. On velocity gradient dynamics and turbulent structure. *Journal of Fluid Mechanics*, 780:60–98, 2015.

[14] Yi Li and Charles Meneveau. Material deformation in a restricted euler model for turbulent flows: analytic solution and numerical tests. *Physics of Fluids*, 19(1), 2007.

[15] Bo Liu, Zhen-Hua Wan, Xi-Yun Lu, and Luo-Qin Liu. A physics-informed deep learning closure for lagrangian velocity gradient evolution. *Physics of Fluids*, 35(11), 2023.

[16] Jesús Martın, César Dopazo, and Luis Valiño. Dynamics of velocity gradient invariants in turbulence: restricted euler and linear diffusion models. *Physics of Fluids*, 10(8):2012–2025, 1998.

[17] Charles Meneveau. Lagrangian dynamics and models of the velocity gradient tensor in turbulent flows. *Annual Review of Fluid Mechanics*, 43:219–245, 2011.

[18] Koji Ohkitani and Seigo Kishiba. Nonlocal nature of vortex stretching in an inviscid fluid. *Physics of Fluids*, 7(2):411–421, 1995.

[19] Sandeep Pandey, Jörg Schumacher, and Katepalli R Sreenivasan. A perspective on machine learning in turbulent flows. *Journal of Turbulence*, 21(9-10):567–584, 2020.

[20] Nishant Parashar, Balaji Srinivasan, and Sawan S Sinha. Modeling the pressure-hessian tensor using deep neural networks. *Physical Review Fluids*, 5(11):114604, 2020.

[21] S. B. Pope. A more general effective-viscosity hypothesis. *Journal of Fluid Mechanics*, 72(2):331–340, 1975.

[22] Stephen B Pope. A more general effective-viscosity hypothesis. *Journal of Fluid Mechanics*, 72(2):331–340, 1975.

[23] Maurits H Silvis, Ronald A Remmerswaal, and Roel Verstappen. Physical consistency of subgrid-scale models for large-eddy simulation of incompressible turbulent flows. *Physics of Fluids*, 29(1), 2017.

[24] Yifeng Tian, Yen Ting Lin, Marian Anghel, and Daniel Livescu. Data-driven learning of mori–zwanzig operators for isotropic turbulence. *Physics of Fluids*, 33(12):125118, 2021.

[25] Yifeng Tian, Daniel Livescu, and Michael Chertkov. Physics-informed machine learning of the lagrangian dynamics of velocity gradient tensor. *Physical Review Fluids*, 6(9):094607, 2021.

[26] P. Vieillefosse. Internal motion of a small element of fluid in an inviscid flow. *Physica A: Statistical Mechanics and its Applications*, 125(1):150–162, 1984.

[27] Michael Wilczek and Charles Meneveau. Pressure hessian and viscous contributions to velocity gradient statistics based on gaussian random fields. *Journal of Fluid Mechanics*, 756:191–225, 2014.

[28] Q.-S. Zheng. On the representations for isotropic vector-valued, symmetric tensor-valued and skew-symmetric tensor-valued functions. *International Journal of Engineering Science*, 31(7):1013–1024, 1993.